

UX Evaluation: Reporting Results

Highlights

- Importance of quality communication and participant anonymity
- Reporting different kinds of data
 - Reporting informal summative results
 - Reporting qualitative results
 - Reporting ecological and emotional impact problems
- Report audiences
- Report content
- Report mechanics
- Report tone

25.1 INTRODUCTION

25.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 25.1). This chapter is about reporting UX evaluation results and applies more or less regardless of the evaluation method or data collection technique and whether data are empirical or analytic.

25.1.2 Importance of Quality Communication

Evaluation reports are often required to communicate across discontinuities of time, location, and people.

- **Time delays.** Re-design activities are often separated from UX evaluation by delays in time that can cause information loss due to human memory limitations.
- **Different people.** This is further aggravated if the people doing the re-design are not the ones who conducted the evaluation. UX evaluation reports with inadequate

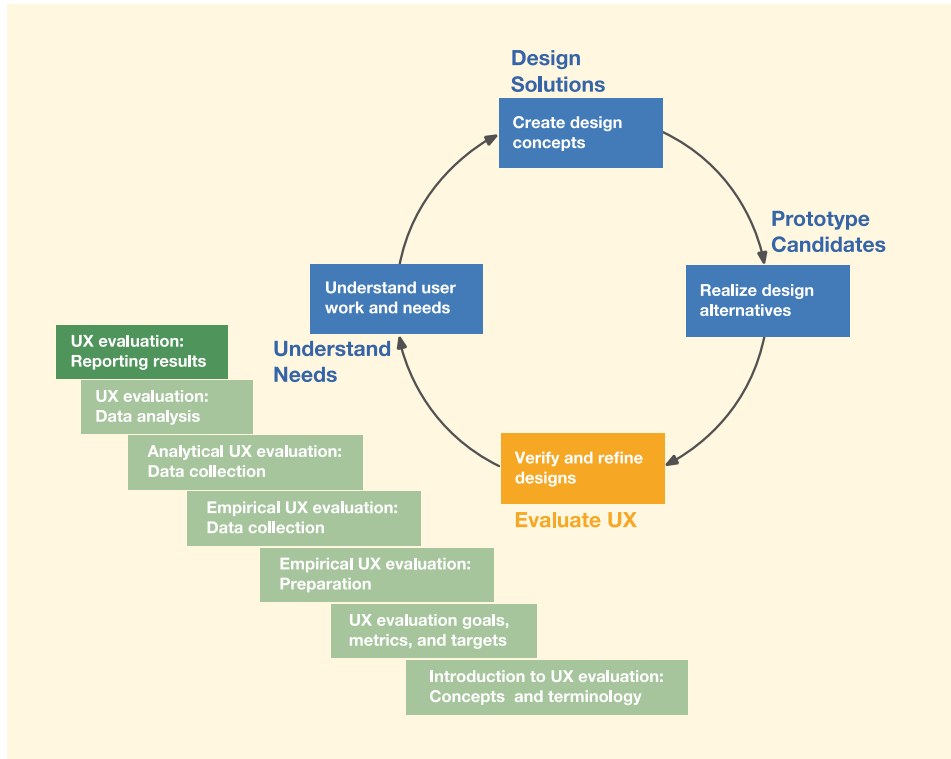


Fig. 25.1

You are here, in the chapter on evaluation data reporting within the evaluate UX activity, in the context of the overall Wheel UX lifecycle.

contextual information or incomplete UX problem descriptions will be too vague for designers who were not present for the evaluation.

- **Different locations.** Evaluation and re-design can occur at different physical locations, rendering all information not well communicated to be unrecoverable.

25.1.3 Participant Anonymity

We remind you, before we get into the details, that regardless of your kind of evaluation or reporting, you must preserve participant anonymity. You should have promised this on your informed consent form, and you have an ethical, and perhaps a legal, obligation to protect it religiously thereafter. The necessity for preserving participant anonymity extends especially to evaluation reporting.

25.1.4 Common Industry Format for Reporting

There is not a lot of UX literature about UX evaluation reporting. One notable exception is the work at the US National Institute of Standards and Technology (NIST) in the early 2000s.

The initial work by NIST in this area was to produce a common industry format (CIF) for reporting formal summative UX evaluation results. Following this initial effort, the group—under the direction of Mary Theofanos, Whitney Quesenbery, and others—organized two workshops in 2005 (Theofanos et al., 2005), and these aimed at a CIF for formative evaluation reports (Quesenbery, 2005; Theofanos & Quesenbery, 2005).

Although their working definition of formative testing was based on having representative users, here we use the slightly broader term of formative evaluation to include usability inspections and other methods for collecting formative usability and user experience data. Their workshop documentation is too detailed to cover here, but parts of this chapter are significantly influenced by that work.

25.1.5 The Need for a Cross-Functional Team in Writing a Report

As we said in [Section 11.2.1](#), design, evaluation, and re-design occur in the context of the pyramid of human user needs. In that section, we showed that cost-importance analysis requires estimating the cost to make changes in each level of the pyramid as well as estimating the importance of making those changes. Estimates of both cost and importance across the pyramid require skills and knowledge from collaborative roles in your cross-functional team, including the UX people, the software engineering (SE) team, business and marketing, and systems engineers—all potentially part of the audience for evaluation reporting. Business people are always included because each business goal translates into a set of ecological, interaction, and emotional concerns.

25.1.6 A Caution About Reporting Summative Results

You are reminded here of your professional responsibilities in reporting summative evaluation results in the context of UX practice. For example, informal summative results are all about engineering and simple averages, gathered from participants recruited without concern for sampling validity and analyzed without concern for statistical significance. Therefore, when you refer to this kind of quantitative UX data in a report, it should never be associated with anything that even remotely looks like a statistical claim.

More recently, emerging formal summative evaluation methods raise new issues about making statistical claims. Customer satisfaction surveys, producing scores generated by surveying a large number of participants using standardized questionnaires, are becoming a common metric in UX practice

these days. These studies assess user satisfaction with products and capabilities. Users are surveyed at a specific cadence (e.g., quarterly or yearly), and the resulting scores are used as a product metric. After each study, the UX people generate a results report that includes validity measures, such as statistical significance. For products with smaller user counts that do not achieve statistical significance, they specify that the quantitative results are not to be relied on, encouraging the reader to focus on the qualitative data alone.

As the UX field is increasingly leaning toward larger amounts of quantitative user and system performance data through surveys, user interface instrumentation to track user performance, and A/B testing, statistical power through inferential statistics is taking on an importance not seen earlier in human–computer interaction or UX evaluation. UX practitioners must now be even more aware of the need to avoid making claims not supported by the statistical significance of the data.

25.1.7 Your Personal Presence in Reporting

Reporting UX evaluation findings is more than just writing it and sending it out. If possible, you should be there to make a presentation when you deliver the report. Your personal presence during reporting can be crucial in getting the payoff for all the resources spent on design and evaluation. Nothing beats face-to-face communication to set the desired tone and expectations. There is no substitute for being there to answer questions and head off costly misunderstandings.

25.2 CONSIDER YOUR AUDIENCE

Theofanos and Quesenbery (2005) stated that choices about content, format, vocabulary, and tone are all about the relationship between report author and audience. The 2005 UPA Workshop Report on formative evaluation reporting (Theofanos et al., 2005) stressed different reporting requirements for different business contexts and audiences.

25.2.1 The Primary Audience: Your Own Team

Your primary audience comprises at least the UX team, the SE team, and the cross-functional team. These people will be making the changes you recommend, and their deep understanding of the problems and potential solutions is essential.

Start with a boilerplate summary of the basics, including evaluation goals, methods, and UX targets and benchmark tasks used. Screen shots illustrating

actual problem encounters are always good for selling your points about problems.

Convey results and product implications clearly and meaningfully to your workmates, informing them about UX flaws in the design and/or informally measured shortcomings in user performance with the purpose of understanding what needs to be done to improve the design in the next iteration. Your report can usually be short and to the point, with little need for embellishments.

25.2.2 Reporting to Inform and/or Influence Management

A management audience needs yet a different kind of report. The team's key goal for reporting to this audience is to influence and convince them that this is part of the process and that the process is working.

- Be concise and get to the point. Begin with an executive summary, which may be the only thing they read. Briefly explain the process, the evaluation goals, the methods, and the UX targets and benchmark tasks used.
- Begin with relevant business goals and how this product/system and UX evaluation relate. Show how the business goals define your evaluation priorities. This is easier if you used UX targets driven by UX goals, based on business and product goals. You need an “explicit connection between the business or test goals and the results” (Theofanos & Quesenbery, 2005; Theofanos et al., 2005).
- Identify the version of the system being evaluated and how it fits overall development context (release schedules, etc.).
- Use cost-importance analysis to focus on UX problems that can be fixed within the number of people-hours allocated in the budget but paint a complete picture of your findings. Screen shots illustrating actual problem encounters may be useful in engaging them in the whole evaluation scene. Include the cost to fix problems identified and motivate with an intuitive estimate of the cost of not fixing various problems.

Relate how these problems impact the overall product portfolio, the evaluation of this particular part of the product, upcoming releases, among other things.

25.2.3 Reporting to Convince Others That Re-design Is Needed

What good is doing the UX evaluation if no one is convinced the problems you found are real, and as a result the design does not get changed? It may be

part of the job of UX practitioners to convince others in the project team to take action about poor UX, as revealed by UX evaluation. This part of the role is especially important in large organizations where people who collect data are not necessarily the same people (or even people who have a close working relationship with them) as those who make the decisions about design changes.

At its extreme, a request to explain why changes are needed could be indicative of an unfortunate management or organizational problem showing a lack of trust that the UX team can do its job. Your response in this report could be what it takes to turn that situation around. Perhaps this is one case where a video recording of users struggling to use a product or system could provide the crucial leverage needed. See the sub-sections on problem impact, recommended fixes, and cost-importance analysis ([Section 25.3.1.3](#)) for further guidance in writing to convince.

25.2.4 Reporting to Customers or Clients

If you are doing the work as a UX consultant, one other kind of audience that may require a different kind of report is your client or customer. If you are not part of the team that created the design you evaluated, then it is best to not start by hitting the client square on with what is wrong with the system.

Explain that finding UX problems in evaluation is simply part of a process focused on fixing problems. Be selective and describe a few examples. Clients and customers will not necessarily want to hear about a whole list of problems with the design of their system. For clients, UX problems are best described with scenarios and screen shots that tell stories of how design flaws affected users, and how your UX engineering process finds and fixes those problems. For your client, focus on patterns and themes that can be fixed and competitive comparisons. What are the industry trends, and how is the existing design aligned with those trends?

Exercise 25.1 UX Evaluation Reporting for Your Product or System

Goal: Write a report of the formative UX evaluation you did on the system of your choice (see [Section 5.6.1.2](#)).

Activities: Report on your informal summative evaluation results using a table showing UX targets, benchmark tasks, questionnaires, and other methods used to gather data, along with target values and observed values.

Add brief statements about whether each UX target was met.

Write a full report on a selected sub-set (about half a dozen) of UX problems found in the qualitative part of your formative UX evaluation. Follow the guidelines in this chapter regarding content, tone, and format, being sure to include re-design proposals for each problem.

Report on the results of your cost-importance analysis, including problem resolutions, for all the problems you reported previously and, if appropriate, some others for context.

Deliverables: Your formative evaluation report.

Schedule: We expect this exercise to take about 2 to 3 hours.

25.3 REPORT GUIDELINES

This section provides different kinds of content and writing styles from which to choose to match your audience.

25.3.1 Problem Reporting

25.3.1.1 Overview

To the project team, the report for an evaluation within an iteration is a re-design proposal. Hornbæk and Frøkjær (2005) show the need for usability evaluation reports that summarize and convey an overview and context, not just lists of problem descriptions by themselves. Your primary audience will get the full details of all the UX problems to be fixed, but the way you organize this presentation will determine its effectiveness.

Begin with an overview and highlights, including:

- Evaluation goals
- Methods used
- UX targets and benchmark tasks used (if appropriate)
- A broad view of the types of UX problems found
- Overall meta-assessment of where everything stands vis-à-vis design, evaluation, re-design iteration.

Overview of the kinds of UX problems can include the following questions:

- Were a large percentage of the problems cosmetic and easy to fix with better labeling and wording?
- Were there many problems that involved user workflow that had to be redesigned?
- Were there deep problems that affected the product's relationship to its ecology?

Lean in to the audience’s abilities to learn visually by using screen shots to illustrate actual problem encounters.

All UX practitioners should be able to write clear and effective evaluation reports, but in their comparative usability evaluation studies, Dumas, Molich, & Jeffries (2004) found that many cannot. They observed a large variation in reporting quality over teams of usability specialists, and most reports were inadequate by their standards.

If you use rapid evaluation methods for data collection, it is especially important to communicate effectively about the analysis and results because this kind of data can otherwise be dismissed easily “as unreliable or inadequate to inform design decisions” (Nayak, Mrazek, & Smith, 1995).

25.3.1.2 Problem reporting: Foundation

As a kind of litmus test for good qualitative problem reporting, to clearly communicate problems and their impact on the system and the business and how to fix those problems, problem reports should be built on a foundation of:

- Specificity—using precise language in problem reports to exactly describe what part of the design was flawed
- Completeness—providing enough context and content detail for stakeholders outside of the evaluation team

25.3.1.3 Problem reporting: Content

Many researchers and practitioners have suggested various content items that might prove useful for problem diagnosis and re-design. The idea is to provide all the essential facts a designer will need to understand and fix the problem. The basic information needed includes the following:

- Problem description
- Best judgment of the causes of the problem in the design
- Estimate of its severity or impact
- Suggested solutions

Problem description. Be sure to describe identified issues as problems rather than solutions. If the problems were experienced by users doing tasks, describe them in that context—users and tasks and the effects of the problems on users. This means

saying, for example, “Users could not figure out what to do next because they did not notice the buttons,” instead of “We need flashing red buttons.”

Estimate of causes. An engineering judgment of the causes of a problem in the UX design is an essential part of diagnosis and perhaps the most important part of the report. Because the flaw in the design is what needs to be fixed, you should connect it with the appropriate design guidelines and/or heuristic violations (see [Chapter 29](#)) as much as possible in terms of UX issues and principles.

Cost to fix. Consult with software engineers on your team and create estimates of the cost to fix each of the major problems. Usually, cost-importance analysis is considered part of the nitty-gritty engineering details that would be beyond the interest or understanding of those outside the UX team and its process. However, cost-importance analysis, especially the prioritization process, can be of interest to those who have to fix the problems and those who have to pay for it.

Impact of problem. Next is a well-reasoned estimate of severity or importance in terms of the impact on users. Potential impact factors include the following:

- Cosmetic problems, while undesirable, will not affect user performance.
- Moderate impact is felt on user task performance and/or satisfaction.
- High expected frequency of occurrence can lead to repeated negative effects on user productivity.
- Major impact on user task performance and/or satisfaction occurs.
- Mission critical; when a user task is blocked, it is a showstopper.

Recommended fix. Help your audience to see the new version of the feature or design and how it will improve user performance and satisfaction, and ultimately your bottom line.

Cost/importance analysis. If it is necessary to justify fixing problems, use cost-importance analysis to make compelling arguments for improved design and a positive impact on users (and therefore on meeting your business goals). Talk about the cost of not fixing certain problems.

25.3.1.4 Prioritize problems to be fixed

It would be nice if we could fix all the UX problems we find in evaluation, but time and budget are always limited. Therefore it is not enough to present lists of UX problems to the primary audience without prioritizing the urgency to fix them.

In [Section 24.4](#), we performed cost-importance analysis with the help of the cross-functional team to establish these priorities. It is now your responsibility to organize your presentation of UX problem descriptions to include this order of priorities in terms of problem criticality, severity, and impact on the business value of the product or system being evaluated, especially if there are showstoppers.

25.3.1.5 Reporting on the ecological and emotional layers of the pyramid

Most of the UX problems you will find and report will occur at the interaction level, but you should also give some attention to the ecological and emotional layers. If there are problems with the design at the ecological level, especially problems with the conceptual design, be sure to stress the importance of this level of design. Inputs from the cross-functional team will help here. Unless the design is successful at the ecological level, evaluation at the other levels will not lead to a good design.

Special discussion should be directed to reporting emotional impact problems because they can be the most important for product improvement and marketing advantage, but these problems and their solutions can also be the most elusive. Emotional impact problems should be flagged as a somewhat different kind of problem with different kinds of recommendations for solutions.

Provide a balanced summary of the overall emotional impact on participants. Report specific positive and negative highlights with examples from particular episodes or incidents. If possible, try to inspire by comparing with products and systems having high emotional impact ratings.

25.3.2 Mechanics of the Report

25.3.2.1 Use a consistent problem-description structure

Establishing your own standards for reporting results helps control broad variation in content, structure, and quality of UX problem reports.

25.3.2.2 Use precision and specificity in your reporting vocabulary

You are communicating with others to accomplish an outcome. To get the audience to share the vision of that outcome or to even understand what outcome you want, you need to communicate effectively.

Sloppy terminology, vague descriptions, and lazy hand-waving are likely to be met with indifference, and the designers and other practitioners are less likely to be interested in the problems and solutions we propose in the report, and they may not take the report seriously.

Instead of saying a dialogue box message text is hard to understand and recommending that someone write it more clearly, you should, in fact, make your own best effort at rewording to clarify the text and say why your version is better. The litmus test criterion for effectiveness is whether the designer who receives your problem report will be able to make better re-design choices (Dumas, Molich, & Jeffries, 2004).

25.3.3 Report Tone and Style

25.3.3.1 Respect feelings in your report tone

The British are too polite to be honest, but the Dutch are too honest to be polite.

—Candid Dutch saying

Begin the report with a positive tone. Most UX professionals realize that they should start off with good things to say about the system being evaluated. However, even when encouraged to be positive, some practitioners in studies (Dumas, Molich, & Jeffries, 2004) proved to be reticent in this regard.

All your audiences deserve respect in evaluation reports. In reporting to customers and clients, most UX professionals appreciate the need to temper reports with restraint. As Dumas, Molich, and Jeffries (2004) stated, “Express your annoyance tactfully.” Even your own team should be addressed with courtesy. The analysis presented is not an assessment of the team or designers, but of the design. Focus on the problem and impact instead of on the people who designed the product.

Your goal is not to get designers angry or resentful, but to get them to act on the report and fix the problems.

Some evaluators believe that design flaws must be stated in strong language to convey the message, but designers say they are insulted by emotional rants: “Being blunt is not helpful; it is simply rude” (Dumas, Molich, & Jeffries, 2004).

25.3.3.2 Avoid jargon

Avoid jargon, acronyms, and esoteric terminology. Consider the breadth of your audience and whether they will know esoteric jargon when they see it.

25.4 SEGUE

In this chapter, we concluded Part 5, our coverage of UX evaluation, with a discussion of reporting. In the next chapter (in Part 6), we switch gears and dig deeper into the agile process discussion we began in [Chapter 4](#) and connect our UX processes with the SE counterparts.